

Fig. 5: Interfacing Arduino with ADXL-335 sensor



Fig. 6: Hand glove

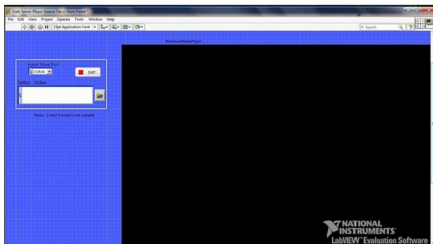


Fig. 7: Front Panel of LabVIEW project

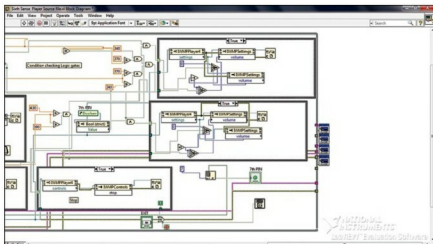


Fig. 8: Block Diagram of LabVIEW project

Arduino

This is an open source physical computing platform based on a simple microcontroller (MCU) board and a development environment for writing software for the board. ATmega328 is used in Arduino UNO board. It has 14 digital input/output (I/O) pins (of which six can be used as PWM outputs), six analogue inputs, a 16MHz ceramic resonator, USB connection, power jack, reset button and an ICSP header. It contains everything needed to support the MCU; simply connect it to a computer with a USB cable or power it with an AC-to-DC adaptor or battery to get started.

In this project, a switch button is used at digital pin 7 of the Arduino board for volume up and down controls. Analogue pins A0, A1 and A2

are used to interface accelerometer ADXL-335. The circuit block diagram is shown in Fig. 5.

LabVIEW with Arduino UNO

Interfacing LabVIEW with Arduino is done through LabVIEW Interface for Arduino (LIFA) application. LIFA allows users to control sensors and acquire data through an Arduino MCU using the graphical programming environment of LabVIEW. By utilising the processing power of LabVIEW running on a desktop computer with I/O and expandability of Arduino platform, users can create many applications.

Tools required include Arduino board, Arduino IDE, LabVIEW, NI-VISA and JKI VIPM.

Steps for interfacing LabVIEW and Arduino UNO board

1. Install LabVIEW 2011 or later (current version 2014). LabVIEW 2014, 32-bit version can run on Windows 7, 64-bit PC.

A full-featured, non-commercial renewable six-month evaluation of LabVIEW can be downloaded from NI's website.

2. Install Arduino IDE. It is free, open source and available for download at www.arduino.cc/en/main/Software. Arduino IDE is required to build LIFA's open source firmware.

3. Install NI-VISA. It provides serial (RS232) drivers for LabVIEW and is available as a free download at www.ni.com/visa/

4. Install JKI VI Package Manager (VIPM). This package manager for LabVIEW provides an easy way to install and update LabVIEW libraries. Download it from jki.net/vipm/download

5. Install LIFA, which is part of the LabVIEW tools network and is available for direct download or using VIPM.

6. Connect Arduino UNO to the host computer. Any Arduino-compatible board should work but LIFA was designed for and tested using